

Lab 3: Logistic Regression

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3.1 Field Goal Success Probability

3.1.1 Data

We have a dataset consisting of field goals, where each row represents a field goal. Each row includes the following variables:

- i : index of the i^{th} field goal
- y_i : 1 if the i^{th} field goal was made, and 0 otherwise
- $yard_i$: the yardline of the i^{th} kick, measured in yards from the opponent's end zone.
- kq_i : kicker quality of the i^{th} kicker.
- $kicker_i$: the name of the kicker for the i^{th} field goal.

3.1.2 Your Task:

1. Model field goal success probability using at least 3 different models, with at least one being a linear regression and one being a logistic regression.
2. For each logistic regression model you fit, report coefficient estimates, standard errors, and 95% confidence intervals. Interpret coefficients on the log-odds scale, and when useful exponentiate them to interpret odds ratios.
3. Use out-of-sample predictive performance to select the best model of the 3. Report the average test log loss for each model, and if you use cross-validation also report the standard error of the test log loss across folds.
4. Write an interpretation of the coefficients of the selected model, making clear both the point estimate and the uncertainty in that estimate.
5. Plot the selected model's predicted make probability against yardline, including a 95% confidence band for the fitted probability. Also compare these fitted probabilities to binned observed make rates.
6. Briefly explain the difference between uncertainty in the fitted make probability and uncertainty in the outcome of one kick. The fitted probability can have a confidence interval, but the actual kick is still Bernoulli: it is either 0 or 1.

3.2 Bradley-Terry NCAA Men's Basketball Power Scores

3.2.1 Data

We have a dataset consisting of the results of NCAA Men's Basketball games, where each row represents a game and includes the following variables:

- i : index of the i^{th} game
- s_i : season of the i^{th} game
- h_i : index of the home team
- a_i : index of the away team
- y_i : 1 if the home team won, and 0 otherwise

3.2.2 Bradley-Terry Model

The Bradley-Terry model supposes each team j has a latent power rating (or strength) β_j and the probability that team j beats team k at home is given by

$$p_{jk} = \text{Logistic}(\beta_0 + \beta_j - \beta_k)$$

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$$p_{jk} = 1 - p_{kj} = \text{Logistic}(\beta_0 + \beta_k - \beta_j)$$

This is similar to the model we set up in Lecture 2, with the addition of the Logistic transformation. Note that β_0 is a home field advantage parameter.

3.2.3 Your Task:

1. Filter the data to include games from the 2023-2024 season (i.e. $s = 2023$)
2. Recode your dataset to create a model dataset like the one in Lecture 3.
3. Fit a Bradley-Terry model to the data, and state what identifiability convention you used for the team ratings, such as centering the ratings to have mean zero or dropping one reference team.
4. Report the estimated coefficients with uncertainty summaries. In particular, visualize team ratings with uncertainty intervals, and for comparisons between two teams focus on uncertainty in the rating difference.
5. Set a Vegas moneyline price for the 2023-2024 NCAA Tournament final between the Purdue Boilermakers and the UConn Huskies (Note: the game is played at a neutral site, so $\beta_0 = 0$). Report:
 - (a) the estimated neutral-site win probability for each team;
 - (b) an approximate 95% confidence interval for that probability;
 - (c) the corresponding moneyline price for the point estimate and for the endpoints of the interval.

6. Hint for the probability interval: if $\delta = \beta_{\text{Purdue}} - \beta_{\text{UConn}}$, then on a neutral court

$$p = \text{Logistic}(\delta) = \frac{1}{1 + e^{-\delta}}.$$

If you build an approximate 95% confidence interval for δ , say $[\delta_L, \delta_U]$, then a corresponding approximate 95% confidence interval for the win probability is

$$\left[\frac{1}{1 + e^{-\delta_L}}, \frac{1}{1 + e^{-\delta_U}} \right].$$

7. Moneyline formula: if a team has win probability p , then the corresponding fair moneyline is

$$\text{Moneyline}(p) = \begin{cases} -100 \cdot \frac{p}{1-p}, & p \geq 0.5 \\ 100 \cdot \frac{1-p}{p}, & p < 0.5 \end{cases}$$

Use this formula for the point estimate and for the endpoints of the probability interval.

8. Briefly explain the difference between uncertainty in the estimated win probability and uncertainty in the actual game outcome.

3.3 A Note on ELO Power Scores

ELO is an **online** or **rolling** version of Bradley-Terry logistic regression power scores, which is updated after every match. These models are frequently used in one-on-one sports, such as chess or tennis.

Let player i 's ELO be β_i . Then the probability that player i beats player j is given by

$$\begin{aligned} p_{ij} = \mathbb{P}(i \text{ beats } j) &= \text{Logistic}(\beta_i - \beta_j) \\ &= \frac{1}{1 + \exp(-(\beta_i - \beta_j))} \end{aligned}$$

Then if player i beats j , update their ELO as follows:

$$\begin{aligned} \beta_i &\leftarrow \beta_i + K \cdot (1 - p_{ij}) \\ \beta_j &\leftarrow \beta_j + K \cdot (0 - p_{ij}) \end{aligned}$$

where K is a constant learning rate we set.